

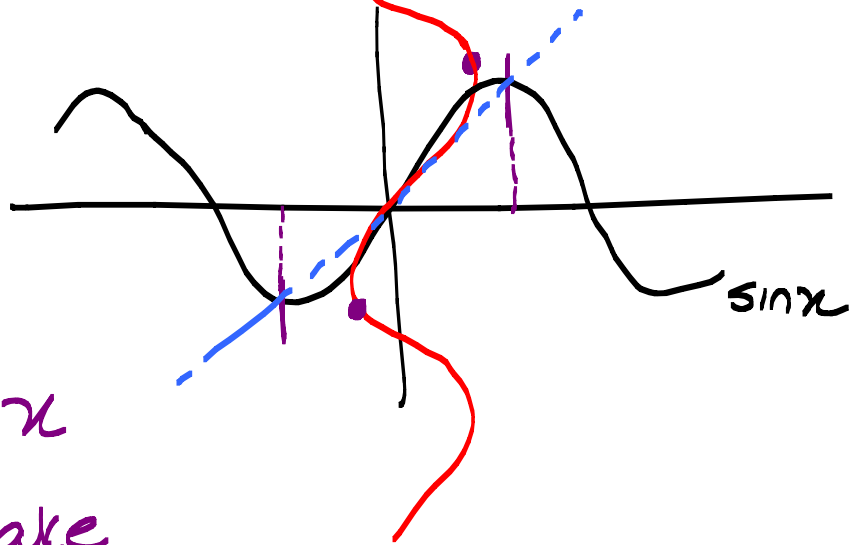
Unit 1.10 Inverse trigonometric functions

Note Title

2014/02/08

pp 63-66

$\sin x$ is not
a one-to-one
function



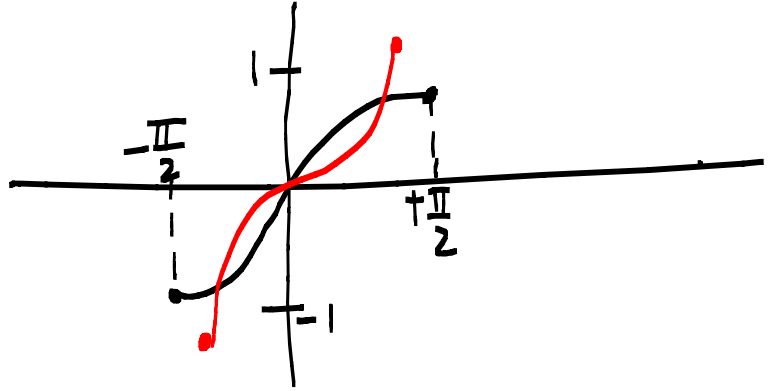
To ensure that $\sin x$
is one-to-one take
 $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Let $y = \sin x$. To find the inverse:
 $x = \sin^{-1} y$
and.

$$\sin^{-1} x = y \iff x = \sin y$$

$x \in [-1, 1]$ $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

To ensure
one to one
function take
 $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$



1. $y = \sin x$

2. $x = \sin y$

3. $\sin^{-1} x = y$ OR
 $\arcsin x = y$

$f(x) = \sin x$

$f^{-1}(x) = \arcsin x$

$D_f = [-\frac{\pi}{2}, \frac{\pi}{2}]$ $D_{f^{-1}} = [-1, 1]$
 $R_f = [-1, 1]$ $R_{f^{-1}} = [-\frac{\pi}{2}, \frac{\pi}{2}]$

$f^{-1}(x) = \arcsin x$

$\sin^{-1} x = y \iff \sin y = x$

$y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$x \in [-1, 1]$

Cancellation equations

$\sin(\arcsin(x)) = x$ if $x \in [-1, 1]$

$\arcsin(\sin(x)) = x$ if $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Example:

Solve $\sin x = -0.5 = -\frac{1}{2}$

$$RA: x = \frac{\pi}{6}$$

$$x = \frac{7\pi}{6} + 2k\pi$$

$$x = \frac{11\pi}{6} + 2k\pi$$

x	x
$k \in \mathbb{Z}$	

Many solutions

Compare that to:

Find $\arcsin(-\frac{1}{2})$

Start with $\arcsin(-\frac{1}{2}) = y$

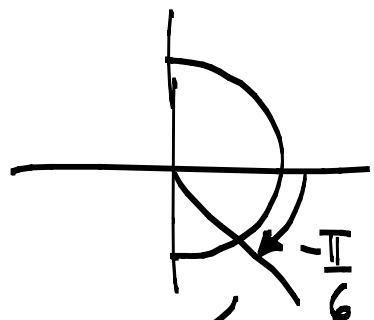
$$\Leftrightarrow \sin(\arcsin(-\frac{1}{2})) = \sin y$$

$$-\frac{1}{2} = \sin y \text{ if}$$

$$[-1, 1]$$

$$[-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\Rightarrow y = -\frac{\pi}{6}$$



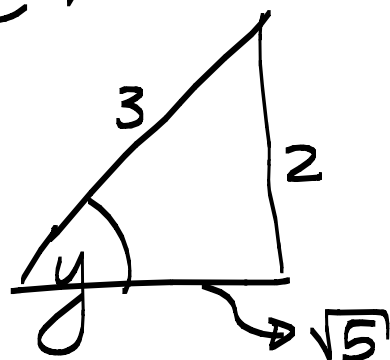
② Determine $\tan(\arcsin \frac{2}{3})$

$$\text{let } \arcsin \frac{2}{3} = y$$

$$\Rightarrow \frac{2}{3} = \sin y$$

$$\in [-1, 1] \checkmark$$

$$y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

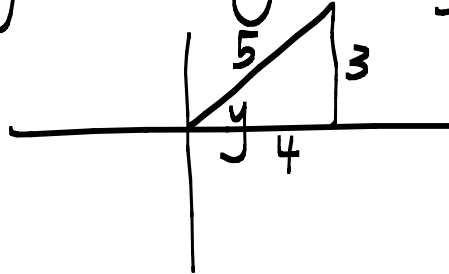


$$\tan(\arcsin \frac{2}{3}) = \tan(y) = \frac{2}{\sqrt{5}}$$

$$(3) \sin \left(2 \sin^{-1} \left(\frac{3}{5} \right) \right)$$

let $\sin^{-1} \left(\frac{3}{5} \right) = y$

$$\Rightarrow \frac{3}{5} = \sin y \text{ and } y \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$



$$\therefore \sin(2y)$$

$$= 2 \sin y \cos y = 2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right) = \frac{24}{25}$$

$$f(x) = \sin x$$

$$D_f = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$R_f = [-1, 1]$$

$$f'(x) = \sin^{-1} x = \arcsin x$$

$$D_{f'} = [-1, 1]$$

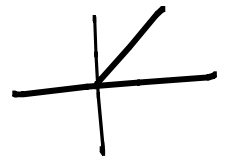
$$R_{f'} = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$(4) (i) \text{ Find } \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \quad (ii) \sin^{-1} \left(-\frac{\sqrt{3}}{2} \right)$$

(i) let $y = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right)$

$$\Rightarrow \sin y = \frac{1}{\sqrt{2}}, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

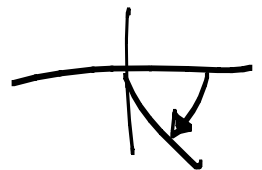
$$\Rightarrow y = \frac{\pi}{4}$$



(ii) let $y = \sin^{-1} \left(-\frac{\sqrt{3}}{2} \right)$

$$\Rightarrow \sin y = -\frac{\sqrt{3}}{2}, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

RA: $\frac{\pi}{3}$ So $y = -\frac{\pi}{3}$



Determine

(i) $\arcsin\left(-\frac{\sqrt{2}}{2}\right)$

$= -\frac{\pi}{4}$

$$\frac{\sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$RA: \frac{\pi}{4}$$

$$\bigcup \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Now we do the same for $\cos x$

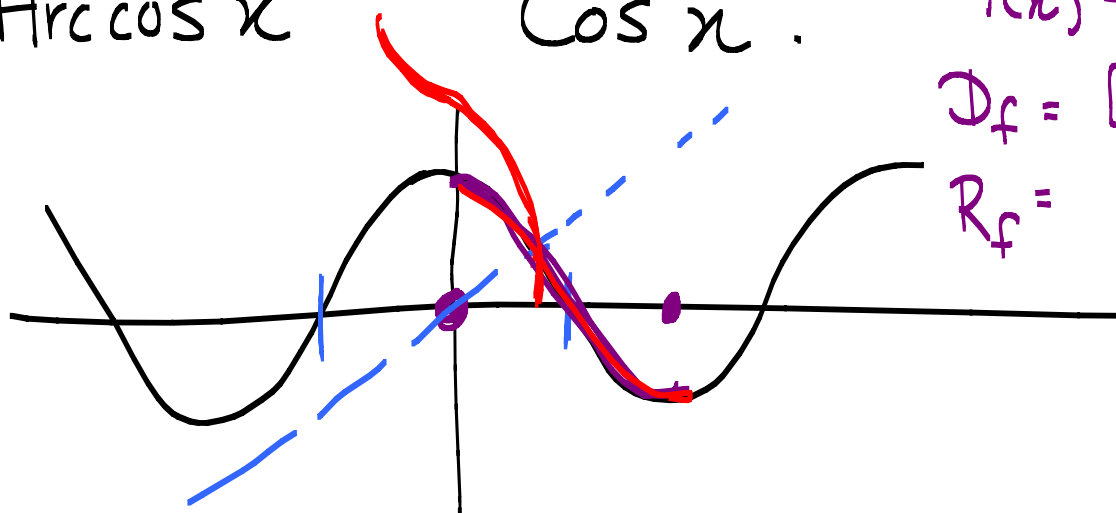
$\arccos x$

$\cos^{-1} x$

$$f(x) = \cos x$$

$$\mathcal{D}_f = [0, \pi]$$

$$\mathcal{R}_f = [-1, 1]$$



1-to-1
function

$$f(x) = \cos x, \quad x \in [0, \pi]$$

$$\mathcal{D}_f = [0, \pi]$$

$$\mathcal{R}_f = [-1, 1]$$

$$f^{-1}(x) = \cos^{-1} x \quad (\arccos x)$$

$$\mathcal{D}_{f^{-1}} = [-1, 1]$$

$$\mathcal{R}_{f^{-1}} = [0, \pi]$$

$$y = \cos^{-1} x$$

$\swarrow$
 $[-1, 1]$

$\Leftrightarrow$

$$\cos y = x$$

$\searrow$
 $[0, \pi]$

As before:

$$\cos(\arccos x) = x \quad \text{if } x \in [-1, 1]$$

$$\arccos(\cos x) = x \quad \text{if } x \in [0, \pi]$$

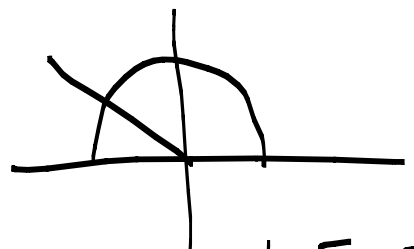
Examples:

1. Find $\arccos\left(-\frac{\sqrt{3}}{2}\right)$

let $y = \arccos\left(-\frac{\sqrt{3}}{2}\right)$

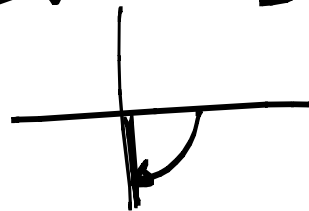
$$\Rightarrow \cos y = -\frac{\sqrt{3}}{2} \quad \text{and } y \in [0, \pi]$$

$$\text{RA: } \frac{\pi}{6} \quad y = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$



2. Find $\arccos(\cos(-\frac{\pi}{2})) \neq -\frac{\pi}{2} \rightarrow \notin [0, \pi]$

let $y = \arccos(0)$
 $= \frac{\pi}{2}$

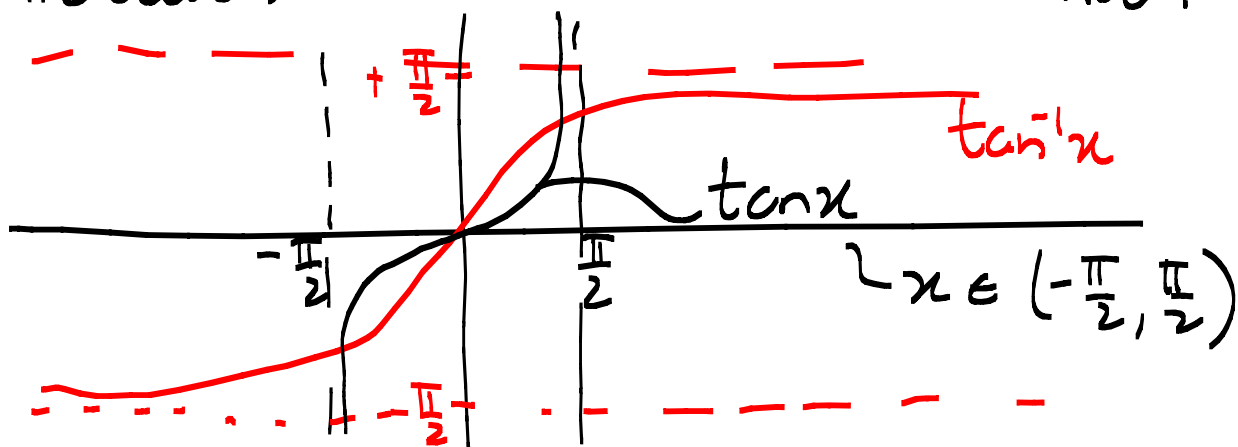


Now same story for $\tan x$

Arctan x

$\tan^{-1} x$

$\tan x$ is
not 1-to-1



Start of with $f(x) = \tan x$ with
 $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ $D_f = (-\frac{\pi}{2}, \frac{\pi}{2})$
 $R_f = \mathbb{R}$

$$y = \arctan x \iff \tan y = x$$

and $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$$f^{-1}(x) = \arctan x \quad D_{f^{-1}} = \mathbb{R}$$

$$R_{f^{-1}} = (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\tan(\arctan x) = x \quad \text{if } x \in \mathbb{R}$$

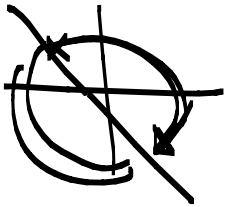
$$\arctan(\tan x) = x \quad \text{if } x \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

Examples:

1. Solve $\tan x = -1$.

RA: $\frac{\pi}{4} \quad \therefore x = -\frac{\pi}{4} + k\pi, \quad k \in \mathbb{Z}$

Multiple answers



2. Find $\arctan(-1)$

Let $y = \arctan(-1)$

$$\Rightarrow \tan y = -1 \quad \text{and } y \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\Rightarrow y = -\frac{\pi}{4} \quad (\text{One answer!})$$

$$3. \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = y$$

$$\Rightarrow \tan y = -\frac{1}{\sqrt{3}} \text{ and } y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$y = -\frac{\pi}{6}$$

$$4. \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = y$$

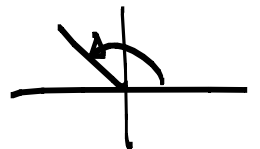
$$\sin y = \frac{\sqrt{3}}{2} \text{ and } y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\therefore y = \frac{\pi}{3}$$

$$5. \cos^{-1}\left(-\frac{1}{2}\right) = y$$

$$-\left(\frac{1}{2}\right) = \cos y \text{ and } y \in [0, \pi]$$

$$y = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$



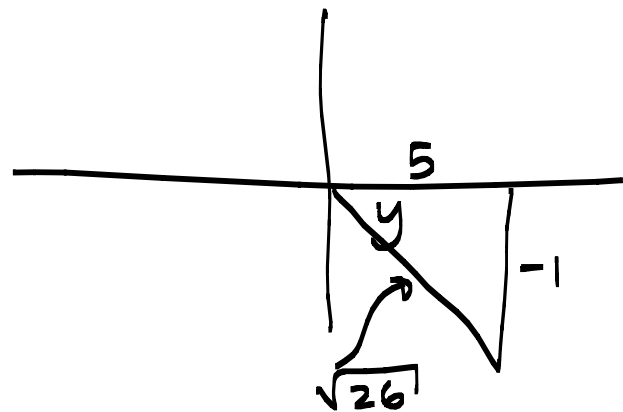
$$6. \text{ Find } \cos(\arctan(-\frac{1}{5}))$$

$$\text{let } y = \arctan(-\frac{1}{5})$$

$$\Rightarrow \tan y = -\frac{1}{5} \text{ and } y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\cos(\arctan(-\frac{1}{5}))$$

$$= \cos y = \frac{5}{\sqrt{26}}$$

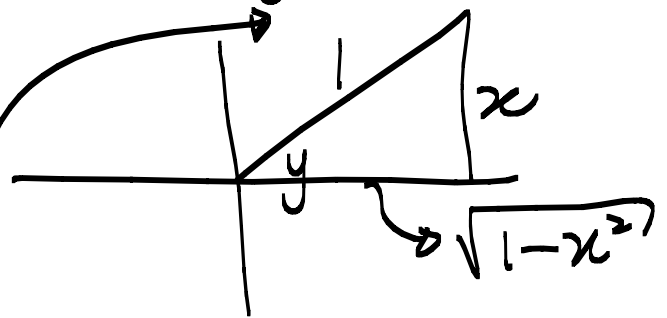


$$7. \text{ Simplify } \tan(\sin^{-1} x)$$

let $y = \sin^{-1} x$

$\Rightarrow \sin y = \frac{x}{1}$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

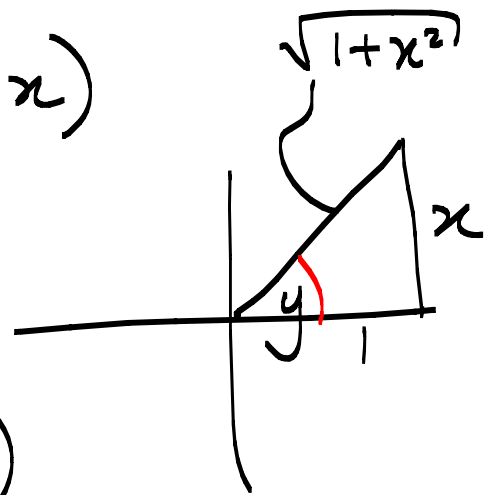
So $\tan(\sin^{-1} x)$
 $= \tan(y) = \frac{x}{\sqrt{1-x^2}}$



8. Simplify $\cos(2 \tan^{-1} x)$

let $y = \tan^{-1} x$

$\Rightarrow \tan y = x$,
 and $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$



So $\cos(2y)$

$= \cos^2 y - \sin^2 y$

$= \left(\frac{1}{\sqrt{1+x^2}}\right)^2 - \left(\frac{x}{\sqrt{1+x^2}}\right)^2$

$= 1 - x^2$ you simplify.

$\cos(y+y)$
 $= \cos y \cos y - \sin y \sin y$

$y \in (-\frac{\pi}{2}, +\frac{\pi}{2})$